



Urban Informatics

Assignment - Week 2

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 16

Total marks: (12 X 1) + (4 X 2) = 20 marks

MCQ/MSQ Question

QUESTION 1: [1 mark]

Which of the following statements about data formats are correct?

- A. CSV is appropriate for structured datasets, such as demographic tables or street-light maintenance records.
 - B. JSON uses name–value pairs enclosed in braces and supports nested objects and arrays.
 - C. XML documents consist of hierarchical elements that describe the meaning of data.
 - D. JSON is more detailed than XML and can be slower to transmit.
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- a. B, C and D
 - b. A, C and D
 - c. A, B and D
 - d. A, B and C

Correct Answer: d. A, B and C

Detailed explanation: Refer to lecture 6. XML is more detailed than JSON and can be slower to transmit.

QUESTION 2: [1 mark]

Which of the following is NOT an example of Database Management System (DBMS)?

- a. Microsoft Access
- b. PostgreSQL
- c. Microsoft Excel
- d. MongoDB

Correct Answer: c. Microsoft Excel

Detailed explanation: Refer to lecture 7.

QUESTION3: [1 mark]

Which of the following are the ACID properties of RDBMS?

- A. atomicity
- B. consistency
- C. insulation
- D. durability



- a. A, B and D
- b. A, C and D
- c. B, C and D
- d. A, B and C

Correct Answer: a. A, B and D

Detailed explanation: Refer to lecture 7.

QUESTION 4: [1 mark]

Which of the following is correct?

- a. Normalization is structuring a relational database in accordance with normal forms to reduce data redundancy and improve data integrity.
- b. Higher normal forms always increase performance
- c. A deletion anomaly occurs when deleting a record intentionally removes other data.
- d. An update anomaly occurs when data is stored in single place.

Correct Answer: a. Normalization is structuring a relational database in accordance with normal forms to reduce data redundancy and improve data integrity.

Detailed explanation: Refer to lecture 8.

QUESTION 5: [2 marks]

Match the following key features with their Database Model.

Database Model	Key features
P. Network model	i. Very fast access, schema-less, highly scalable
Q. Relational model	ii. Structured schema, data integrity, SQL, normalization
R. Object-oriented model	iii. Many-to-many relationships, graph-like links
S. Key-Value model	iv. Complex data types, inheritance, similar to object-oriented programming

- a. P-ii; Q-i; R-iii; S-iv
- b. P-i; Q-iii; R-iv; S-ii
- c. P-iii; Q-ii; R-iv; S-i
- d. P-ii; Q-iii; R-i; S-iv

Correct Answer: c. P-iii; Q-ii; R-iv; S-i

Detailed explanation: Refer to lecture 7.

QUESTION 6: [1 mark]

Which statement best explains why unstructured data poses a greater analytical challenge for urban planning compared to structured data?

- a. It lacks a predefined schema, requiring advanced NLP, computer vision, and machine learning for feature extraction and pattern recognition.



- b. It contains only qualitative information that cannot be quantified for spatial analysis.
- c. It cannot be stored in relational databases and must always be converted into raster formats.
- d. It is always smaller in volume but higher in redundancy than structured datasets.

Correct Answer: a. It lacks a predefined schema, requiring advanced NLP, computer vision, and machine learning for feature extraction and pattern recognition.

Detailed Solution: Refer to lecture 6, last slide.

QUESTION 7: [1 mark]

Third Normal Form automatically satisfies _____.

- a. 2NF
- b. BCNF
- c. 4NF
- d. 5NF

Correct Answer: a. 2NF

Detailed Solution: Refer to lecture 8.

QUESTION 8: [2 mark]

Match the following-

Database Model	Key features
P. Candidate Key	i. May include extra attributes beyond the minimum needed for uniqueness.
Q. Primary Key	ii. A set of attributes that can uniquely identify each record in a table, and a table may have more than one such key.
R. Superkey	iii. An attribute that links one table to the primary key of another table to maintain valid relationships.
S. Foreign Key	iv. The selected candidate key used as the main unique identifier for records and for defining relationships.

- a. P-ii; Q-i; R-iii; S-iv
- b. P-i; Q-iii; R-iv; S-ii
- c. P-iii; Q-ii; R-i; S-iv
- d. P-ii; Q-iv; R-i; S-iii

Correct Answer: d. P-ii; Q-iv; R-i; S-iii

Detailed explanation: Refer to lecture 9.

QUESTION 9: [1 mark]

NoSQL databases are primarily chosen when:



- a. The application requires strict ACID transactions
- b. The system needs to scale horizontally across distributed servers
- c. The data schema must remain fixed and highly normalized
- d. The goal is to reduce all data to a single unified table for faster queries

Correct Answer: b. The system needs to scale horizontally across distributed servers

Detailed Solution: Refer to lecture 10.

QUESTION 10: [1 mark]

A _____ is a functional dependency in which $X \rightarrow Z$ (X determines Z) indirectly, by virtue of $X \rightarrow Y$ and $Y \rightarrow Z$ (where it is not the case that $Y \rightarrow X$).

- a. transitive dependency
- b. partial dependency
- c. multi-valued dependency
- d. join dependency

Correct Answer: a. transitive dependency

Detailed Solution: Refer to lecture 9.

QUESTION 11: [1 mark]

Which of the following statements are true about normalization of databases?

- A. Normalization is the process of organizing tables in a relational database to reduce redundancy and avoid update, insertion, and deletion anomalies.
 - B. Higher levels of normalization can be achieved even without satisfying the conditions of the previous normal forms.
 - C. It is based on a series of rules called normal forms, which ensure that each fact is stored in one appropriate place only.
 - D. Normalization improves data integrity and consistency by eliminating duplicate and dependent data.
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- a. A, B and D
 - b. A, C and D
 - c. B, C and D
 - d. A, B and C

Correct Answer: b. A, C and D

Detailed Solution: Refer to lecture 8.

QUESTION 12: [2 marks]

Match the type of NoSQL databases with their example-

NoSQL database	Examples
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P. Wide-Column Databases	i. Neo4j
Q. Graph Databases	ii. Redis
R. Document Databases	iii. HBase
S. Key–Value Databases	iv. MongoDB

- a. P-iv; Q-ii; R-iii; S-i
- b. P-iii; Q-i; R-iv; S-ii
- c. P-iv; Q-iii; R-ii; S-i
- d. P-ii; Q-i; R-iv; S-iii

Correct Answer: b. P-iii; Q-i; R-iv; S-ii

Detailed Solution: Refer to lecture 10.

QUESTION 13: [1 mark]

_____ is important because they provide a secure interface that enables applications to retrieve, share, and modify database information without giving direct access to the database.

- a. API
- b. Non-Relational Database
- c. Normalization
- d. DBMS

Correct Answer: a. API

Detailed Solution: Refer to lecture 10. APIs are essential because they act as a secure bridge that allows applications to access, exchange, and update database data without directly opening or exposing the database itself.

QUESTION 14: [2 marks]

Match the normal forms with their characteristics-

Normal Forms	Characteristics
P. First Normal Form (1NF)	i. Eliminates transitive dependency
Q. Third Normal Form (3NF)	ii. Removes multi-valued dependencies
R. Boyce & Codd Normal Form (BCNF)	iii. Ensures each determinant is a candidate key
S. Fourth Normal Form (4NF)	iv. Eliminates repeating groups and ensures atomic values

- a. P-iv; Q-i; R-iii; S-ii
- b. P-iii; Q-ii; R-i; S-iv
- c. P-ii; Q-i; R-iii; S-iv
- d. P-iv; Q-iii; R-i; S-ii

Correct Answer: a. P-iv; Q-i; R-iii; S-ii

Detailed explanation: Refer to lecture 9.



QUESTION 15: [1 mark]

_____ is a relational database management system that combines the Jet database engine with a user-friendly graphical interface.

- a. Oracle Database
- b. Microsoft Access
- c. MariaDB
- d. PostgreSQL

Correct Answer: b. Microsoft Access

Detailed Solution: Refer to lecture 7.

QUESTION 16: [1 mark]

What is the BASE property of NoSQL databases?

- a. Batch Access, Static State, Error-free Consistency
- b. Balanced Access, Structured Storage, Encrypted Consistency
- c. Backup Availability, Stable Schema, Exact Consistency
- d. Basically Available, Soft State, Eventually Consistent

Correct Answer: d. Basically Available, Soft State, Eventually Consistent

Detailed Solution: Refer to lecture 10.

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